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Unified hierarchical contrastive learning for video captioning

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ABSTRACT

Video captioning involves the generation of textual descriptions for videos. Many video captioning models are typically trained using video-text pairs and adopt maximum likelihood estimation as their learning method, which overlooks caption distinctiveness and results in high resemblance in model-generated captions. Although contrastive methods improve distinctiveness, their performance is influenced by the reference model, and they rely on a two-stage training process: first training a reference model, then training a target model to surpass the reference. In this work, we propose a simple yet effective Unified Hierarchical Contrastive Learning (UHCL) method, which harnesses triamese decoders and hierarchical contrastive learning to enhance overall performance and encourage distinctiveness within a unified framework. Specifically, UHCL employs triamese decoders to compute contrastive loss using additional matched and mismatched video-caption pairs. Furthermore, adaptive token fusion module is utilized to reduce the redundancy of visual tokens, guiding the model to apply hierarchical contrastive constraints at the noun and verb levels. It is worth noting that UHCL does not require prior information, thus avoiding the risk of introducing noise, and only a single decoder is required during inference, with no additional computational overhead. Extensive experiments conducted on MSR-VTT and MSVD, demonstrate the effectiveness of the proposed method and its superiority over the state-of-the-art methods. Particularly, UHCL outperforms the best results by improving 7.4 % and 0.6 % absolute CIDEr scores on the MSVD and MSR-VTT datasets, respectively. The code will be available at https://github.com/sunhaoying97/UHCL

1. Introduction

Video captioning automatically describes video content using natural language through machine learning. It has potential applications in various fields, such as disability assistance, human-computer interaction, and automated commentary [1–4].

Existing deep learning-based video captioning approaches are predominantly based on encoder-decoder architectures [5–10] which are trained using the video-text pairs. It usually results in generated captions that highly resemble the training set [11] and lack distinctiveness [12]. Two kinds of approaches have been proposed to address the aforementioned issue. The first one is prior information-based methods, as shown in Fig. 1(a), which input the extra the prior information, such as concepts [13–15], retrievals [16–18], and topics [19], into the decoder to enhance the detail, diversity of descriptions, and distinctiveness to some extent. However, the inaccurate prior information possibly influences the quality of the generated captions. The second one is contrastive-based methods, [12,20], as shown in Fig. 1(b). These methods employ a pre-trained reference model as guidance, it can improve the distinctiveness of the target model. Nonetheless, the performance is influenced by the reference model : they fail to account for relationships among positive samples, and in real application scenarios, the reference model is updated [12] multiple times to maintain its effectiveness, which incurs a high cost.

In fact, the reference model is implicitly derived from the training data [12]. In other words, the guidance provided by the reference model is dissolved within the training set itself, which inherently contains both similar and different samples. Contrastive learning [21], commonly used in self-supervised learning [22–24], is particularly well-suited for modeling such relationships, as it pulls positive sample pairs closer while pushing negative pairs apart. Therefore, it is feasible to train a unified framework that reduces intra-class distances among positive samples while increasing inter-class distances among negative samples through contrastive learning [19]. This allows the model to simultaneously capture both the distinctiveness and similarity inherent in the data at a relatively low computational cost.

In this paper, we propose a simple yet effective Unified Hierarchical Contrast Learning (UHCL) method for video captioning, as illustrated in Fig. 1(c). During training, a video and its corresponding caption (anchor), positive sample (caption from the same video but different from the anchor) and negative sample (caption from different videos) are input into triamese decoders. Subsequently, the relationships among the

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generated video captions are constrained using contrastive learning. To further constrain the solution space, visual tokens are processed twice through the adaptive token fusion module. The fused visual tokens are then subsequently employed to compute contrastive loss hierarchically at both the noun and verb levels. Notably, while triamese decoders are used during training, only a single decoder is used during inference to generate sentence, thereby avoiding additional computational overhead.

The contributions of this paper are summarized below:

- A novel, simple yet effective contrastive framework for video captioning, named Unified Hierarchical Contrastive Learning (UHCL), is proposed. Leveraging triamese decoders and adaptive token fusion modules, this framework enforces hierarchical contrastive learning constraints without requiring additional models or training.
- The relationships between the generated video captions and those of the anchor, positive samples, and negative samples are constrained using hierarchical contrastive learning. This method applies to both noun and verb levels through the adaptive token fusion module, enabling multi-granularity constraints during model training.
- Experiments demonstrate that the proposed method outperforms state-of-the-art methods by 7.4 % and 0.6 % in absolute CIDEr scores on the widely used MSVD and MSR-VTT datasets, respectively. Additionally, it surpasses baseline methods by 9.7 % and 4.4 % in absolute CIDEr scores, respectively.

2. Related work

In this section, we provide a brief review of video captioning methods based on deep learning, followed by an overview of research on contrastive learning and its applications in captioning.

2.1. Video captioning

Recent encoder-decoder-based video captioning methods have demonstrated significant performance improvements and can be mainly categorized into two types. The first type is trained in an end-to-end manner [9,15,25] without relying on pre-trained feature extractors for visual feature extraction. Typically, the visual encoder is pre-trained on large datasets [26,27] and then the entire model fine-tuned on video captioning datasets. While effective, this end-to-end approach generally involves a large number of parameters. The second type employs frozen pre-trained feature extractors to obtain video features with some methods further extracting motion features [28–32], and the model is subsequently trained on video captioning datasets. Improvements in this type include enhancing modality alignment capabilities [32–34], optimizing network structures, and employing advanced feature extractors [26] to boost feature extraction capabilities [10]. A notable advancement in this approach is the extraction and integration of additional prior information, such as concepts [13–15] retrievals [16–18], optical flow [9], sentiment [35], object detection [28–30], and topics [19]. Although these methods effectively extract prior information from videos, they could potentially mislead the model [14]. Furthermore, the training of these methods typically employs maximum likelihood estimation (MLE), without accounting for the relationships between captions from different videos or varying captions within the same video, leading to a lack of diversity and distinctiveness [12]. Our approach falls into the second category but differs in key aspects. By leveraging triamese decoders and contrastive learning, we account for multi-level relationships between captions from different videos and varying captions within the same video. Moreover, our method achieves remarkable performance without relying on prior information, thereby eliminating the introduction of noise and additional inference overhead.

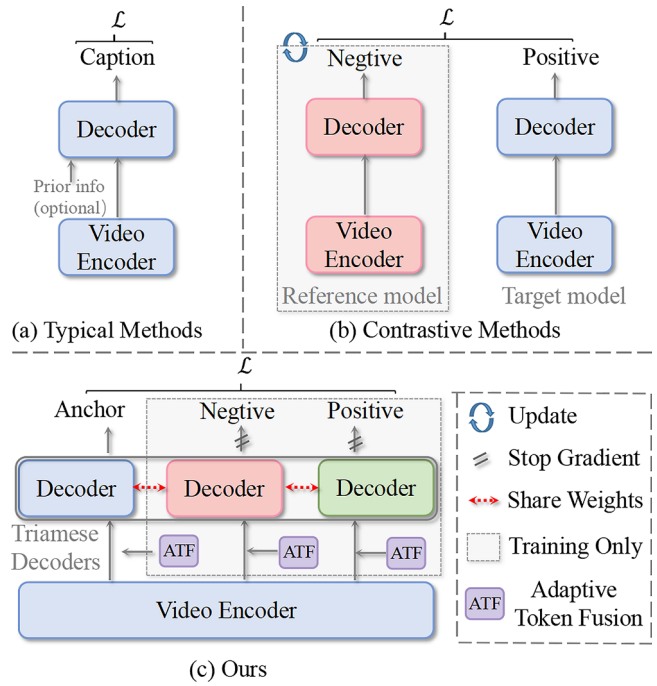

Fig. 1. A training pipeline comparison between our method and previous methods. (a) Typical methods condition on visual features and prior information to maximize the probability of corresponding video captions, without considering the relationship between mismatched video-caption pairs. Although effective, these methods still encounter several issues, including high resemblance and a lack of distinctiveness in the model-generated captions. (b) Contrastive methods are influenced by the reference model and require a two-stage training: first training a reference model, then using matched and mismatched visual-text pairs to train the target model to outperform the reference model. (c) Our simple yet effective one-stage UHCL framework harnesses triamese decoders and adaptive token fusion modules. Notably, it does not require prior knowledge or a reference model and introduces no additional inference overhead.

2.2. Contrastive learning

The core idea of contrastive learning [21] is to bring positive pairs closer together while pushing negative pairs further apart [22]. This approach is inherently well-suited for modeling relationships between anchors, positive and negative samples/features, and has been widely applied in self-supervised learning [22–24,36–38], image-to-image translation [39], image dehazing [40] and image-language pre-training [26]. To alleviate the limitations of MLE, which often results in high resemblance [11] and a lack of distinctiveness [12] in model-generated captions, contrastive learning [12] integrates negative samples into the loss calculation, effectively mitigating issues of indistinctiveness in image captions. Building on this foundation, contrastive learning [20] has been extended to video captioning, enhancing caption distinctiveness and diversity by mining the hardest negative samples. Despite its effectiveness, these methods typically rely on a two-stage training process: first training a reference model and then training a target model to surpass the reference. Unlike the aforementioned methods [12,20] that depend on pre-trained reference models, our approach achieves a unified framework through triamese decoders, which dynamically learn the relationships between mismatched and matched video-text pairs. Additionally, fine-grained relationships within captions, such as those involving nouns and verbs, are explicitly constrained. Notably, inference can be performed with single-stage training, ensuring the model both straightforward and efficient.

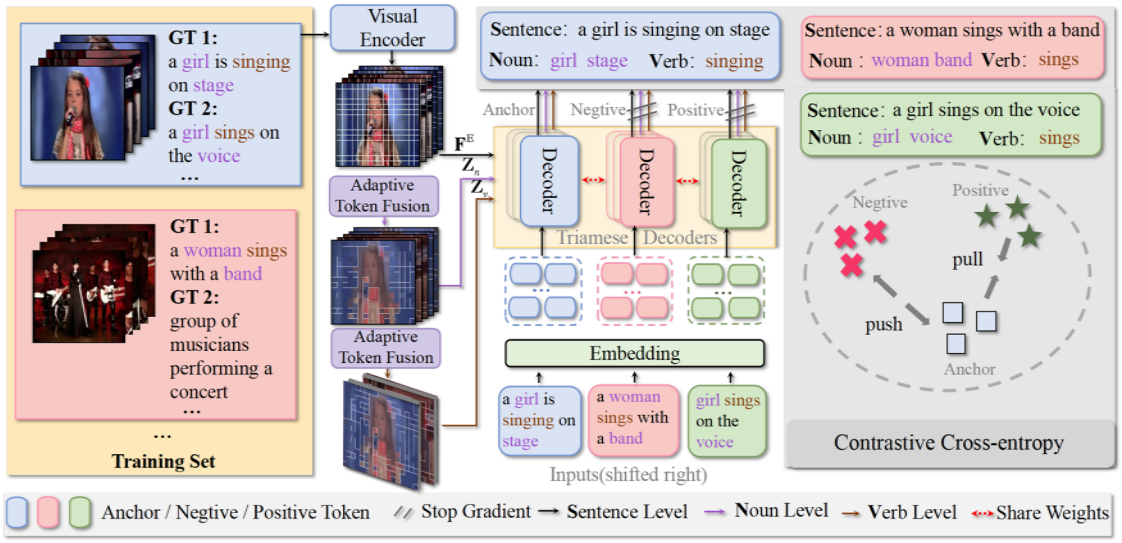

Fig. 2. Overview of the proposed UHCL. First, the encoded video tokens are additionally processed through the adaptive token fusion module during training to facilitate subsequent noun- and verb-level constraints. Then, additional captions from the same and different videos are randomly sampled from the training set as positive and negative examples. These captions, along with the processed visual tokens, are then fed into the triamese decoders to generate the corresponding text. Finally, hierarchical contrastive learning is employed to optimize the relationships among different samples at three levels: sentence, noun, and verb. During inference, only a single decoder is required, incurring no additional computational overhead. Best viewed in color.

3. Methodology

Fig. 2 illustrates the proposed Unified Hierarchical Contrast Learning (UHCL) framework for video captioning. Built upon an encoder-decoder architecture, UHCL is designed to capture multi-level relationships between captions of different videos or varying captions of the same video within a unified framework, thereby enhancing the distinctiveness and quality of the generated captions. In the following, we first present a baseline for video captioning, followed by a detailed explanation of the adaptive token fusion (ATF) module and hierarchical contrastive learning approach. Finally, the training and inference phases of our method are described.

3.1. Video captioning baseline

Without loss of generality, we introduce a Transformer baseline here. Take the input sampled video frames as $\varsigma = \{\varsigma_1, \varsigma_2, \dots \varsigma_M\}$ of length M, input frames are encoded by the pre-trained backbone(e.g. CLIP's image encoder [26]). Each frame is mapped to d_h (e.g. 512d) visual feature, resulting $M \times d_h$ dynamic features $\mathbf{F}_\mathrm{B} = \mathrm{Backbone}(\varsigma)$ for each video, d_h is the backbone dimension. We use the output of backbone, a sequence of frames embedding, as our visual representation. The baseline comprises a Transformer-based video encoder and decoder, the video features $\mathbf{F}_\mathrm{B}$ are input into a Transformer video encoder ($\mathbb{E}$) to obtain the enhanced features $\mathbf{F}_\mathrm{E} = \mathbb{E}(\mathbf{F}_\mathrm{B})$, $\mathbf{F}_\mathrm{E} \in \mathbb{R}^{M \times D}$, where D (e.g. 768d) represents the predefined dimensionality of the video encoder. The enhanced features are then fed into a Transformer decoder ($\mathbb{D}$) to generate caption $s = \mathbb{D}(\mathbf{F}_\mathrm{E})$ for each video. The model is trained using cross-entropy loss:

$$\mathcal{L}_{CE}(s, \varsigma) = -\sum_{t=1}^{T_{\max}} \mathbb{I}(s_t) \log p_\theta(s_t | s_{:t-1}; \varsigma) \tag{1}$$

where $T_{\max}$ is the max length of caption sentence, p is the probability of predicted word, θ is the trainable parameters of the model, s_t is the sentence which has generated at the tth time step, and $\mathbb{I}(s_t) \in \mathbb{R}^{|\mathcal{V}|}$ is the one-hot encoding of s_t, where the value equals to 1 only at the position t, $|\mathcal{V}|$ is the length of vocabulary $\mathcal{V}$.

3.2. Adaptive token fusion (ATF)

To facilitate hierarchical contrastive learning at the noun and verb levels without increasing inference time or affecting sentence decoding, a straightforward strategy involves spatiotemporally fusing visual tokens to distinguish them from the original visual tokens used for sentence decoding. Empirical observations indicate that the number of nouns in a sentence typically exceeds the number of verbs, while both are fewer than the total number of words in the sentence. To this end, adaptive token fusion (ATF) is introduced to fuse visual tokens, reduce redundant information, and sequentially assist noun- and verb-level constraints. This approach infers the decoding target through the differences and lengths of visual tokens without modifying the decoder. It not only reduces computational overhead but also optimizes the feature space through multi-level constraints. The implementation details of the ATF module are shown in Fig. 3.

Specifically, taking the first fusion of visual tokens as an example, let $\mathbf{F}_\mathrm{E} = \{f_i^\mathrm{E}\}_{i=1}^M$, $\mathbf{F}_\mathrm{E} \in \mathbb{R}^{M \times D}$, f_i^E represent the ith frame-level visual features obtained from the Transformer visual encoder $\mathbb{E}(\cdot)$, which serve as the input to the ATF module. Here, M is the length of sampled frames. Initially, a one-dimensional convolution layer followed by LayerNorm is applied to obtain the enhanced features $\tilde{\mathbf{F}}_\mathrm{E} = \{\tilde{f}_i^\mathrm{E}\}_{i=1}^M$. Then, DPC-KNN [41], a density peak clustering algorithm based on K-nearest neighbors, is utilized for clustering visual tokens. Subsequently, the local density ρ_i for each token is computed based on its K-nearest neighbors, as defined by the following formula:

$$\rho_i = \exp\left(-\frac{1}{K} \sum_{\tilde{f}_k^\mathrm{E} \in \mathrm{KNN}(\tilde{f}_i^\mathrm{E}, \tilde{\mathbf{F}}_\mathrm{E})} \left\| \tilde{f}_k^\mathrm{E} - \tilde{f}_i^\mathrm{E} \right\|^2\right) \tag{2}$$

where $\mathrm{KNN}(\tilde{f}_i^\mathrm{E}, \tilde{\mathbf{F}}_\mathrm{E})$ denotes the K-nearest neighbors of $\tilde{f}_i^\mathrm{E}$ in $\tilde{\mathbf{F}}_\mathrm{E} \backslash \tilde{f}_i^\mathrm{E}$. The notation $\tilde{\mathbf{F}}_\mathrm{E} \backslash \tilde{f}_i^\mathrm{E}$ indicates the removal of $\tilde{f}_i^\mathrm{E}$ from $\tilde{\mathbf{F}}_\mathrm{E}$. Intuitively, ρ_i represents the local density of token $\tilde{f}_i^\mathrm{E}$. Following this, the distance

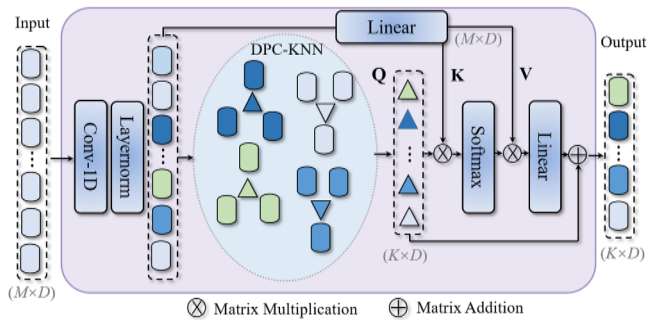

Fig. 3. Adaptive token fusion (ATF) module.

metric δ_i for token $\tilde{f}_i^{\mathrm{E}}$ is computed as follows:

$$\delta_i = \begin{cases} \min\limits_{j:\rho_j>\rho_i} ||\tilde{f}_j^{\mathrm{E}} - \tilde{f}_i^{\mathrm{E}}||^2, if \exists j \text{s.t.} \rho_j > \rho_i. \\ \max\limits_{j} ||\tilde{f}_j^{\mathrm{E}} - \tilde{f}_i^{\mathrm{E}}||^2, otherwise. \end{cases} \tag{3}$$

where δ represents the distance from $\tilde{f}_i^{\mathrm{E}}$ to other high-density tokens. These relatively high values $\rho_i * \delta_i$ are designated as cluster centers. Next, other tokens are assigned to their nearest cluster centers based on Euclidean distance. Inspired by [42] and [43], the average token of each cluster is used to represent the corresponding cluster. Subsequently, we use the clustered tokens as $\mathbf{Q} \in \mathbb{R}^{K\times D}$, and the original enhanced features $\tilde{\mathbf{F}}_{\mathrm{E}} \in \mathbb{R}^{M\times D}$ as $\mathbf{K}$ and $\mathbf{V}$, feeding them into the attention module. Finally, the fused output $\mathbf{Z} \in \mathbb{R}^{K\times D}$ is obtained through a linear layer and a residual connection.

For subsequent fusions, $\mathbf{Z}$ is used as input with an adjusted K value, and the same procedure is repeated. Unlike [42], the ATF module merges visual tokens to generate tokens for nouns and verbs, explicitly utilizes hierarchical contrastive learning to finely constrain noun and verb information within the tokens.

The impact of alternative token fusion methods on our approach was also examined, including non-parametric clustering (e.g., DPC-KNN [41] alone) and neural network-based non-clustering token fusion (e.g., MLP). Performance comparisons are presented in ablation studies (refer to Section 4.3).

3.3. Hierarchical contrastive learning

Our hierarchical approach differs from methods that incorporate generated nouns and verbs embeddings [44,45] or semantic priors [14,15], which are directly fed into the decoder for sentence generation. Such methods may introduce noise and require the decoding process to pause until the extraction of prior information is complete. In contrast, our unified hierarchical contrastive learning framework imposes fine-grained constraints on the solution space during training by harnessing triamese decoders and adaptive token fusion, effectively eliminating the reliance on prior information and reference models.

Hierarchical contrastive learning involves randomly selecting a positive and a negative sample from the training set (details provided below). These samples are fed into triamese decoders, and the loss is computed through hierarchical contrastive learning across sentence, noun, and verb levels. The ATF module, described in the previous section, is employed to generate noun- and verb-level visual tokens for fine-grained constraints. A comprehensive explanation is provided below.

1) Sentence Level Contrastive Learning

In contrastive learning, two essential aspects must be addressed: the careful selection of positive and negative samples, which determines the quality of feature discrimination, and the definition of an appropriate feature or metric space, which governs the embedding structure and similarity measurement.

For the selection of positive and negative samples, the characteristics of video captioning datasets allow direct sampling from the training set. Specifically, the video-text pairs in the training set are represented as $\{(\varsigma^a, s^b)\}$, $a \in N_\varsigma$, $b \in N_a$, Let N_ς denotes the number of training videos and N_a is the number of captions corresponding for the video ς^a. A positive caption is randomly selected from the same video ς^a, denotes as s^{b^+} (abbreviated s^+), $b^+ \neq b$, $b^+ \in N_a$, while a negative caption is randomly chosen from a different video, denoted as s^{b^-} (abbreviated s^-), $b^- \notin N_a$.

For the metric space, we directly adopt the cross-entropy $\mathcal{L}_{CE}$ mentioned in Eq. (1). The sentence-level cross-entropy losses for the anchor, positive, and negative samples are denoted as $\mathcal{L}_S^{Anchor} = \mathcal{L}_{CE}(s, \varsigma)$, $\mathcal{L}_S^{Pos} = \mathcal{L}_{CE}(s^+, \varsigma)$ and $\mathcal{L}_S^{Pos} = \mathcal{L}_{CE}(s^-, \varsigma)$, respectively.

To enable contrastive learning within a unified framework, we design triamese decoders to process the ground truth (anchor), positive, and negative samples. The triamese decoder consists of three decoders with shared weights, facilitating the decoding of positive and negative samples. For different levels of decoding, each level is equipped with a distinct classification head to prevent information leakage and minimize interference between levels.To mitigate the influence of positive and negative samples on anchor decoding, gradient flow is truncated during their decoding. This approach was found to stabilize model training and improve computational efficiency. During inference, a single-head decoder is employed to directly generate the target caption, incurring no additional computational overhead. Therefore, with the triamese decoders, the sentence-level contrastive cross-entropy loss can be computed directly and is formulated as follows:

$$\begin{aligned} \mathcal{L}_{CL}^S &= \frac{\mathcal{L}_S^{Anchor}(s,\varsigma) + \mathcal{L}_S^{Pos}(s^+,\varsigma)}{\mathcal{L}_S^{Neg}(s^-,\varsigma)+\varepsilon} \\ &= -\frac{\sum_{t=1}^{T_{\max}} \mathbb{I}(s_t) \log p_\theta(s_t|s_{:t-1};\varsigma) + \sum_{t=1}^{T_{\max}} \mathbb{I}(s_t^+) \log p_\theta(s_t^+|s_{:t-1}^+;\varsigma)}{\sum_{t=1}^{T_{\max}} \mathbb{I}(s_t^-) \log p_\theta(s_t^-|s_{:t-1}^-;\varsigma)+\varepsilon} \end{aligned} \tag{4}$$

where $\varepsilon = 1e-7$ is added to prevent the denominator from being zero, and $\mathbb{I}(s_t) \in \mathbb{R}^{|\mathcal{V}|}$ is the one-hot encoding of s_t, where the value equals to 1 only at the position t, $|\mathcal{V}|$ is the length of vocabulary $\mathcal{V}$. $\mathcal{L}_{CL}^S$ not only considers the relationship between the generated caption and anchor but also ensures that the anchor and positive samples are closer in the embedding space, while the anchor and negative samples are farther apart. Based on the above, we have achieved contrastive learning-based video caption generation within a unified framework, enhancing the model's distinctiveness without the need for reference model.

2) Noun Level Contrastive Learning

To further optimize the solution space and achieve fine-grained contrastive constraints while reusing the triamese decoders, we employ the adaptive token fusion (ATF) module. This module not only eliminates redundancy in visual tokens but also infers the decoding target through the differences and lengths of visual tokens without modifying the decoder, enabling the triamese decoders to operate at the desired level and enhancing model performance without adding extra inference overhead.

For the visual input, we utilize $\mathbf{Z}_n = \mathrm{ATF}(\mathbf{F}_{\mathrm{E}})$, $\mathbf{Z}_n \in \mathbb{R}^{K_n\times D}$, which is the result of the ATF fusion described in Section 3.2. Here, K_n denotes the length of the visual token $\mathbf{Z}_n$ used for noun-level constraints after ATF fusion, and $\mathbf{F}_{\mathrm{E}}$ represents the output of the visual encoder.

For the textual input, given an anchor sentence s, the corresponding target nouns $s^n = \{s_{t_n}^n\}_{t_n=1}^{T_{\max}^n}$ is constructed as follows:

$$s_{t_n}^n = s_t if\, POS(s_t) \in \{noun\} \tag{5}$$

where POS(·) denotes the part of speech of a word s_t, $T_{\max}^n$ denotes the length of nouns.

In fine-grained supervision at the noun and verb levels, it is assumed that dependencies among words are relatively weak, focusing primarily on direct alignment between individual words and visual features. Therefore, unlike sentence-level contrastive learning, causal encoding is not required for fine-grained losses. Instead, they are conditioned solely on the input video without relying on contextual information from the preceding target words. This design simplifies the computational logic of

the loss while emphasizing local alignment with visual features, thereby enhancing the effectiveness of fine-grained constraints for nouns and verbs. Therefore, the loss function for the noun-level constraint in anchor is defined as:

$$\mathcal{L}_N^{Anchor}(s^n, \mathbf{Z}_n) = -\sum_{t_n=1}^{T_{\max}^n} \mathbb{I}(s_{t_n}^n) \log p_\theta(s_{t_n}^n | \mathbf{Z}_n) \tag{6}$$

where $\mathbb{I}$ is the one-hot encoding. Similarly, the loss functions for nouns in the positive and negative samples are defined as follows:

$$\mathcal{L}_N^{Pos}(s^{n+}, \mathbf{Z}_n) = -\sum_{t_n=1}^{T_{\max}^n} \mathbb{I}(s_{t_n}^{n+}) \log p_\theta(s_{t_n}^{n+} | \mathbf{Z}_n) \tag{7}$$

$$\mathcal{L}_N^{Neg}(s^{n-}, \mathbf{Z}_n) = -\sum_{t_n=1}^{T_{\max}^n} \mathbb{I}(s_{t_n}^{n-}) \log p_\theta(s_{t_n}^{n-} | \mathbf{Z}_n) \tag{8}$$

where $s_{t_n}^{n+}$ and $s_{t_n}^{n-}$ represent nouns in the positive sample s^+ and negative sample s^-, respectively, which are constructed in a manner similar to Eq. (5). Additionally, $\mathbb{I}$ is the one-hot encoding. Thus, our noun-level contrastive cross-entropy loss is given by:

$$\begin{aligned}\mathcal{L}_{CL}^N &= \frac{\mathcal{L}_N^{Anchor}(s^n, \mathbf{Z}_n) + \mathcal{L}_N^{Pos}(s^{n+}, \mathbf{Z}_n)}{\mathcal{L}_N^{Neg}(s^{n-}, \mathbf{Z}_n) + \varepsilon} \\ &= -\frac{\sum_{t_n=1}^{T_{\max}^n} \mathbb{I}(s_{t_n}^n) \log p_\theta(s_{t_n}^n | \mathbf{Z}_n) + \sum_{t_n=1}^{T_{\max}^n} \mathbb{I}(s_{t_n}^{n+}) \log p_\theta(s_{t_n}^{n+} | \mathbf{Z}_n)}{\sum_{t_n=1}^{T_{\max}^n} \mathbb{I}(s_{t_n}^{n-}) \log p_\theta(s_{t_n}^{n-} | \mathbf{Z}_n) + \varepsilon}\end{aligned} \tag{9}$$

where $\varepsilon = 1e-7$ is added to prevent the denominator from being zero.

3) Verb Level Contrastive Learning

Similar to noun-level contrastive learning, for the visual input, we leverage ATF module for further fusion based on $\mathbf{Z}_n$, resulting in $\mathbf{Z}_v = \mathrm{ATF}(\mathbf{Z}_n)$, $\mathbf{Z}_v \in \mathbb{R}^{K_v \times D}$, where K_v denotes the length of the fused visual token $\mathbf{Z}_v$ used for verb-level constraints.

For the textual input, given an anchor sentence s,the corresponding target verbs $s^v = \{s_{t_v}^v\}_{t_v=1}^{T_{\max}^v}$ is constructed as:

$$s_{t_v}^v = s_t if\, POS(s_t) \in \{verb\} \tag{10}$$

where POS(·) denotes the part of speech of a word s_t, $T_{\max}^v$ denotes the length of verbs.

Therefore, the loss function for the verb-level constraints in anchor is defined as:

$$\mathcal{L}_V^{Anchor}(s^v, \mathbf{Z}_v) = -\sum_{t_v=1}^{T_{\max}^v} \mathbb{I}(s_{t_v}^v) \log p_\theta(s_{t_v}^v | \mathbf{Z}_v) \tag{11}$$

where $\mathbb{I}$ is the one-hot encoding. Similarly, analogous to Eqs. (7) and (8), the loss functions for the verb-level positive samples s^{v+} and the verb-level negative samples s^{v-} are represented as $\mathcal{L}_V^{Pos}$ and $\mathcal{L}_V^{Neg}$, respectively, with their detailed formulations provided below.

$$\mathcal{L}_V^{Pos}(s^{v+}, \mathbf{Z}_v) = -\sum_{t_v=1}^{T_{\max}^v} \mathbb{I}(s_{t_v}^{v+}) \log p_\theta(s_{t_v}^{v+} | \mathbf{Z}_v) \tag{12}$$

$$\mathcal{L}_V^{Neg}(s^{v-}, \mathbf{Z}_v) = -\sum_{t_v=1}^{T_{\max}^v} \mathbb{I}(s_{t_v}^{v-}) \log p_\theta(s_{t_v}^{v-} | \mathbf{Z}_v) \tag{13}$$

where $\mathbb{I}$ is the one-hot encoding, s^{v+} and s^{v-} are constructed from the positive sample s^+ and negative sample s^-, respectively, in a manner similar to Eq. (10). Finally, similar to Eq. (9), the verb-level contrastive cross-entropy loss is computed as follows:

$$\begin{aligned}\mathcal{L}_{CL}^V &= \frac{\mathcal{L}_N^{Anchor}(s^v, \mathbf{Z}_v) + \mathcal{L}_N^{Pos}(s^{v+}, \mathbf{Z}_v)}{\mathcal{L}_N^{Neg}(s^{v-}, \mathbf{Z}_v) + \varepsilon} \\ &= -\frac{\sum_{t_v=1}^{T_{\max}^v} \mathbb{I}(s_{t_v}^v) \log p_\theta(s_{t_v}^v | \mathbf{Z}_v) + \sum_{t_v=1}^{T_{\max}^v} \mathbb{I}(s_{t_v}^{v+}) \log p_\theta(s_{t_v}^{v+} | \mathbf{Z}_v)}{\sum_{t_v=1}^{T_{\max}^v} \mathbb{I}(s_{t_v}^{v-}) \log p_\theta(s_{t_v}^{v-} | \mathbf{Z}_v) + \varepsilon}\end{aligned} \tag{14}$$

where $\mathbb{I}$ is the one-hot encoding, $\varepsilon = 1e-7$ is added to prevent the denominator from being zero.

3.4. Model training and inference

1) Training: The UHCL model calculates contrastive losses at the sentence, noun, and verb levels. The total training loss of the model is defined as follows:

$$\mathcal{L} = \mathcal{L}_{CL}^S + \mathcal{L}_{CL}^N + \mathcal{L}_{CL}^V \tag{15}$$

UHCL employs triamese decoders, consisting of three weight-sharing decoders. During training, to prevent model instability, stop-gradient operations are applied when computing the decoding losses for positive and negative samples.

2) Inference: During the inference stage, UHCL employs a single decoder to perform anchor sentence decoding without requiring verbs or nouns. Thanks to the design of our network, UHCL does not rely on a reference model, ensuring that its performance is not constrained by such dependencies. Additionally, our model avoids reliance on extra prior information, thereby eliminating potential noise and the overhead associated with extracting prior information. Compared to similar methods like C4C [10], our approach does not increase inference overhead while significantly enhancing performance (referred to Section 4.5).

4. Exeriment

This section begins by describing the experimental setup. Next, UHCL is compared with state-of-the-art methods and qualitatively analyzed. Then, ablation studies are conducted to assess the effectiveness and efficiency of the UHCL architecture. Subsequently, self-retrieval experiments are performed to validate the model's ability to enhance caption distinctiveness, and model efficiency is then evaluated through comparisons with baseline approach. Finally, the visualization results, case studies, single caption scenarios and limitations are presented in sequence.

4.1. Experimental setup

1) Dataset and Evaluation Metrics: Experiments were conducted on two widely used datasets: MSR-VTT [46] and MSVD [47]. The MSVD dataset consists of 1970 videos, each annotated with approximately 40 captions. The average video duration is around 10 s. Following the protocols in [14,46], 1200 videos were used for training, 100 for validation, and 670 for testing. The MSR-VTT dataset is a general video captioning dataset comprising 10,000 video clips, each annotated with 20 captions. The average duration of each clip is approximately 15 s. We followed the official split [46], using 6153 clips for training, 497 clips for validation, and 2090 clips for testing. To evaluate and compare our method against previous approaches, we adopted four widely used evaluation metrics: BLEU@4 (B@4) [48], a precision-based metric; METEOR (M) [49] , which calculates sentence-level alignment scores; ROUGE (R) [50]which measures the longest common subsequence between predicted and reference captions; CIDEr (C) [51], a consensus-based metric.

2) Parameter Settings: The maximum number of words in the generated caption was set to $T_{\max} = 48$, with a maximum of $T_{\max}^n = 5$ nouns and $T_{\max}^v = 3$ verbs. The length of visual tokens after the first ATF fusion is $K_n = 5$, and after the second ATF fusion, it is $K_v = 3$. The model was trained using a batch size of 32 for a total of 25 epochs with the Adam optimizer [52]. For bias, LayerNorm.bias and LayerNorm.weight, weight decay was not applied; for other parameters, an L2 weight decay of 0.01 was used. The learning rate was initialized to 7e-6 with a warm-up proportion of 0.1, where the learning rate linearly increased during the warm-up period and then linearly decayed to zero. During evaluation/inference, the beam search size was set to 5.

3) Implementation Details: For each video, we uniformly sampled $M = 20$ frames. A 12-layer ViT-B/32 with a patch size of 32 [53] (abbreviated as $\mathrm{CLIP}_{\mathrm{ViT-B/32}}$) was used to extract features. Specifically, following [10], a pre-trained $\mathrm{CLIP}_{\mathrm{ViT-B/32}}$ [26] with 12 layers and the patch size 32 after video-text matching pre-training [10] was employed as the

Table 1
Quantitative results on MSVD dataset. †denotes reproduced results. IRN-V2, RN, 3D-RNX and CLIP are short for INCEPTION-RESNET-V2 [64], RESNET [65], 3D RESNEXT [66] and $\text{CLIP}_{\text{ViT-B/32}}$ [26], respectively. Ret denotes retrieval.

Method	Year	Feature		Ret	Detection	MSVD			
		Extractor	Motion			B@4 ↑	R ↑	M ↑	C ↑
STG-KD [28]	2020	RN-101	I3D	/	Object	52.2	73.9	36.9	93.0
ORG-TRL [29]	2020	IRN-v2	3D-RNX	/	Object	54.3	73.9	36.4	95.2
LSRT [30]	2022	IRN-v2	I3D	/	Object	55.6	73.5	37.1	98.5
STA-FG [54]	2020	RN-152	C3D	/	Concept	52.7	/	34.5	/
MAD-SAP [55]	2020	RN-152	3D-RNX	/	Concept	53.3	72.0	35.4	90.8
TTA [32]	2021	RN-152	3D-RNX	/	Concept	51.8	72.4	35.5	87.7
R-ConvED [18]	2022	IRN-v2	/	✓	/	50.5	/	34.4	83.4
SGN [56]	2021	RN-101	3D-RNX	/	/	52.8	72.9	35.5	94.3
MGCMP [57]	2021	IRN-v2	3D-RNX	/	/	55.8	74.5	36.9	98.5
C4C†[10]	2021	CLIP	/	/	/	56.3	75.9	38.7	104.9
HRNAT [58]	2022	IRN-v2	I3D	/	/	55.7	74.1	36.8	98.1
TVRD [59]	2022	IRN-v2	C3D	/	/	50.5	71.7	34.5	84.3
RSFD [60]	2023	RN-101	3D-RNX	/	/	51.2	72.9	35.7	96.7
CARE [14]	2023	CLIP	3D-RNX	✓	Concept	56.3	75.6	39.1	106.9
VEIN [35]	2024	CLIP	3D-RNX	/	/	55.7	74.4	37.6	98.9
EvCap [61]	2024	RN-101	I3D	/	Object	53.6	74.3	36.7	107.2
MAN [62]	2024	RN	3D-RNX	/	/	59.7	74.3	36.2	101.5
RLHMN [44]	2024	IRN-v2	C3D	/	Object	59.9	74.3	36.2	104.7
STGN [63]	2025	RN-101	C3D	/	Object	**61.5**	75.6	37.7	102.8
Ours	/	CLIP	/	/	/	59.3	**77.5**	**40.5**	**114.6**

Table 2
Quantitative results on MSR-VTT dataset. IRN-V2: INCEPTION-RESNET-V2, RN: RESNET, 3D-RNX: 3D RESNEXT, CLIP: $\text{CLIP}_{\text{ViT-B/32}}$. Ret and A denotes retrieval and audio information, respectively.

Method	Year	Feature			Ret	Detection	MSR-VTT			
		Extractor	A	Motion			B@4 ↑	R ↑	M ↑	C ↑
STG-KD [28]	2020	RN-101	/	I3D	/	Object	40.5	60.9	28.3	47.1
ORG-TRL [29]	2020	IRN-v2	/	3D-RNX	/	Object	43.6	62.1	28.8	50.9
LSRT [30]	2022	IRN-v2	/	I3D	/	Object	42.6	61.0	28.3	49.5
STA-FG [54]	2020	RN-152	/	C3D	/	Concept	40.8	/	27.4	/
MAD-SAP [55]	2020	RN-152	/	3D-RNX	/	Concept	41.3	61.4	28.3	48.5
TTA [32]	2021	RN-152	/	3D-RNX	/	Concept	41.4	61.1	27.7	46.7
OpenBook [17]	2022	IRN-v2	/	/	✓	/	43.1	61.9	29.0	52.3
R-ConvED [18]	2022	IRN-v2	/	/	✓	/	37.8	/	27.6	48.3
SGN [56]	2021	RN-101	/	3D-RNX	/	/	40.8	60.8	28.3	49.5
MGCMP [57]	2021	IRN-v2	/	3D-RNX	/	/	41.7	62.1	28.9	51.4
C4C [10]	2021	CLIP	/	/	/	/	46.1	63.7	30.7	57.7
HRNAT [58]	2022	IRN-v2	/	I3D	/	/	42.1	61.6	28.0	48.2
TVRD [59]	2022	IRN-v2	/	C3D	/	/	43.0	62.2	28.7	51.8
RSFD [60]	2023	RN-101	/	3D-RNX	/	/	43.4	62.3	29.3	53.1
CARE [14]	2023	CLIP	✓	3D-RNX	✓	Concept	48.7	65.2	31.5	59.7
VEIN [35]	2024	CLIP	/	3D-RNX	/	/	44.1	62.6	30.0	55.3
EvCap [61]	2024	RN-101	/	I3D	/	Object	45.5	64.5	30.9	53.8
MAN [62]	2024	RN	/	3D-RNX	/	/	41.3	61.4	28.0	49.8
RLHMN [44]	2024	IRN-v2	/	C3D	/	Object	45.1	63.6	28.8	54.2
STGN [63]	2025	RN-101	/	C3D	/	Object	44.9	61.5	31.2	61.5
Ours	/	CLIP	/	/	/	/	**49.4**	**65.6**	31.7	61.7
Ours_2	/	CLIP	/	/	/	/	49.2	**65.6**	**31.8**	**62.1**

Table 3
Ablation study on the components of hierarchical contrastive learning.

Exp.	Model	MSR-VTT				MSVD			
		B@4 ↑	R ↑	M ↑	C ↑	B@4 ↑	R ↑	M ↑	C ↑
1	Baseline	46.1	63.7	30.7	57.7	56.3	75.9	38.7	104.9
2	$+\mathcal{L}_{CL}^{S}$	47.8	65.2	31.5	60.3	58.5	77.1	40.2	112.3
3	$+\mathcal{L}_{CL}^{S}+\mathcal{L}_{CL}^{N}$	48.1	65.0	31.7	61.4	58.2	77.2	40.3	112.6
4	$+\mathcal{L}_{CL}^{S}+\mathcal{L}_{CL}^{V}$	48.8	65.5	31.6	61.1	58.8	77.2	40.2	113.0
5	$+\mathcal{L}_{CL}^{S}+\mathcal{L}_{CL}^{N}+\mathcal{L}_{CL}^{V}$	**49.4**	**65.6**	**31.7**	**61.7**	**59.3**	**77.5**	**40.5**	**114.6**

Table 4
Ablation experiment of the contrastive loss calculation.

Exp.	Contrastive loss method	MSVD			
		B@4 ↑	R ↑	M ↑	C ↑
1	HCL [20]	49.5	/	31.7	57.7
2	Subtraction	58.7	**77.6**	**40.5**	112.3
3	Division (ours)	**59.3**	77.5	**40.5**	**114.6**

backbone to extract rich features. The pre-trained CLIP has proven effective for video captioning tasks [10,13,14]. The extracted video features were fed into a two-layer Transformer video encoder with a hidden layer dimension of $D = 768$ and 12 attention heads to obtain enhanced visual features. These were then passed to a two-layer Transformer decoder with the same hidden size to generate captions. A dropout probability of 0.1 was applied across all attention and other layers. The NLTK toolkit (https://github.com/nltk/nltk) was used to determine word attributes within the captions, enabling the extraction of verbs and nouns.

Table 5
Ablation study of the token fusion scheme.

Exp.	Token fusion method	MSVD			
		B@4 ↑	R ↑	M ↑	C ↑
1	DPC-KNN [41]	59.0	76.9	39.6	110.7
2	MLP	58.5	77.2	**40.6**	112.3
3	ATF (ours)	**59.3**	**77.5**	40.5	**114.6**

Table 6
Ablation study on stop gradient and weight sharing.

Exp.	**w/o** stop gradient	**w/o** share weights	MSVD			
			B@4 ↑	R ↑	M ↑	C ↑
1	✓		56.4	75.8	38.9	104.6
2		✓	54.1	74.3	37.7	98.3
3			**59.3**	**77.5**	**40.5**	**114.6**

We manually annotated one-tenth of the MSVD training set for part-of-speech tagging and estimated that the NLTK toolkit achieves over 99 % accuracy. Consequently, only a small number of verbs and nouns were mislabeled, which is unlikely to materially affect captioning performance. Using NLTK, we found that only 18.26 % of captions in the MSVD dataset contain adjectives and 4.63 % contain adverbs. These proportions are relatively small. Therefore, this study primarily employs nouns and verbs for hierarchical contrastive learning.

4.2. Comparison with state-of-the-art methods

To evaluate the performance of UHCL, we compared it with the following four categories of state-of-the-art video captioning methods using video features extracted by pre-trained models: (1) Methods with detection: EvCap [61], RLHMN [44], STGN [63], STG-KD [28], ORG-TRL [29], STA-FG [54], MAD-SAP [55], LSRT [30], and TTA [32]; (2) Retrieval-enhanced: OpenBook [17], R-ConvED [18], and CARE [14]; (3) Methods only with motion: SGN [56], MGCMP [57], HRNAT [58], TVRD [59], RSFD [60], and VEIN [35]; (4) Method without prior information: C4C [10], and MAN [62]. Table 1 and 2 exhibit the experimental results of all methods on the MSVD and MSR-VTT databases, respectively.

1) MSVD: As presented in Table 1, Our method surpasses other approaches, including those that leverage CLIP for feature extraction, such as CARE and VEIN, across all evaluation metrics. Notably, both our method and C4C operate without relying on additional information, in contrast to other approaches that incorporate extra data, such as optical flow, detection, motion, and retrieval. Despite this, our approach consistently achieves superior performance compared to the majority of methods, except for a few that slightly outperform it on the B@4 metric.

2) MSR-VTT: As listed in Table 2, our method consistently outperformed all comparison methods. For instance, CARE leveraged multimodal information to enhance concept priors and mitigate noise, thereby improving caption quality. In contrast, our method avoided the overhead of using extensive multimodal purification priors and the noise introduced by prior information, demonstrating the simplicity and efficiency of our UHCL. Additionally, the relatively small gain on MSR-VTT compared with MSVD may be attributable to the use of just one negative sample per anchor, a quantity that could be insufficient for large datasets. To verify this hypothesis, we added an additional negative sample, using one positive and two negative samples during training (Ours_2). The results indicate that most metrics improve when two negative samples (Ours_2) are used compared with only one (Ours), demonstrating that incorporating additional negative samples during training can enhance performance on larger datasets.

4.3. Ablation studies

To thoroughly analyze the proposed method, various ablation studies were conducted on the MSVD or MSR-VTT datasets. These studies included the hierarchical contrastive learning loss (Table 3), the contrastive learning scheme (Table 4), the hierarchical contrastive token fusion scheme (Table 5) and weight sharing and gradient Stopping (Table 6).

1) Hierarchical Contrastive Learning Loss

As shown in Table 3, ablation experiments were conducted on the contrastive learning components (Eqs. (4), (9) and (14)). C4C [10] was used as our baseline. Exp. 2 in Table 3 demonstrates that our sentence-level contrastive learning significantly improves all metrics, highlighting its effectiveness in video captioning. The metrics showed slightly improvements when incorporating contrastive learning at the noun and verb levels (Exp.2 and Exp.3) compared to sentence-level contrastive learning alone. This modest enhancement may be attributed to the fact that sentence-level learning inherently captures some aspects of nouns and verbs but remains insufficient. Hierarchical contrastive learning effectively addresses this limitation.

2) Contrastive Learning Scheme

As illustrated in Table 4, various methods for calculating contrastive loss were compared. Subtraction (Exp. 2) refers to the method of calculating contrastive loss by subtracting the negative sample loss. HCL [20] is a hard contrastive learning method for video captioning that requires mining the hardest mismatched pairs and a reference model. In contrast, our approach integrates hierarchical contrastive learning with triamese decoders within a unified framework. Compared to HCL, our method (Exp. 3) demonstrates an absolute improvement of 56.9 % in CIDEr scores.

3) Token Fusion Scheme

To demonstrate the effectiveness of the adaptive token fusion module, ablation experiments were conducted on different fusion methods, as shown in Table 5. These included parameter-free clustering token fusion (removing the neural network component of adaptive token fusion module and retaining DPC-KNN, Exp. 1) and MLP fusion (Exp. 2). Our adaptive token fusion (ATF) module (Exp. 3) exhibited superior performance. This improvement is likely attributable to the module's ability to integrate the combined advantages of neural networks and clustering. It not only captures semantic information within the data but also reduces computational complexity and enhances feature representation capabilities, offering greater flexibility.

4) Weight Sharing and Gradient Stopping

During training, it was observed that stop gradient and weight sharing contribute to training stability and performance enhancement (Exp. 1 and Exp. 2 in Table 6). The stop-gradient technique mitigates interference from positive and negative samples, thereby enhancing training stability and minimizing gradient conflicts. Weight sharing can improve model performance by enhancing data utilization, consistency, and stability through the learning of diverse samples. Additionally, weight sharing reduces model complexity and overfitting, thereby further improving overall performance. As illustrated in Fig. 4(a), we visualized the impact of stop gradient and weight sharing operations on the training process. The results indicate that our method achieves superior performance on the validation set more rapidly during training, demonstrating that the incorporation of stop gradient and weight sharing facilitates faster model convergence. Additionally, our method demonstrates a shorter training time. We present the differences in average training time per step for each configuration across epochs, as shown in Fig. 4(b). The average training time per step of our model decreases with each epoch, highlighting the efficiency of our method's design.

4.4. Self-retrieval

Distinctive captions are defined by their ability to uniquely match a specific video among a collection of videos. Intuitively, such captions are likely to encapsulate the key and unique content of the video,

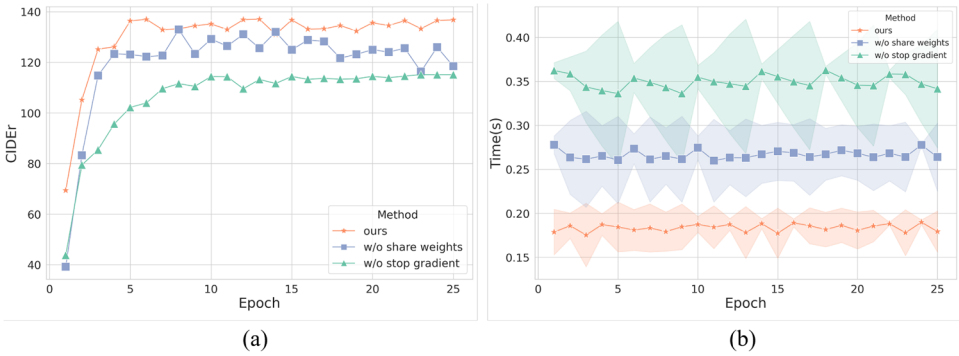

Fig. 4. (a) Comparison of CIDEr scores for various techniques on the validation set during training on the MSVD dataset. (b) Average time consumed per step in each epoch for different techniques during MSVD training.

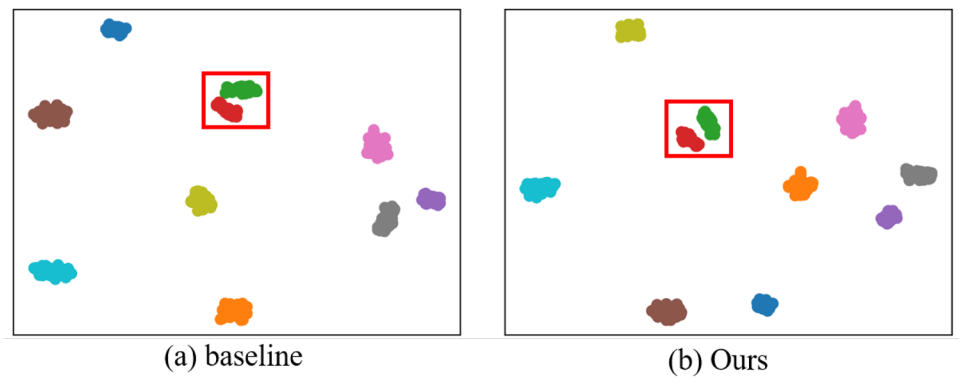

Fig. 5. T-SNE visualization [67] of text embeddings produced by (a) baseline and (b) our UHCL.

making it easier to retrieve the original video when the caption is provided. To evaluate the effect of model design on the distinctiveness of generated captions, we employ a self-retrieval approach as described in [12]. Specifically, we use the MSVD test set comprising 670 videos $(\varsigma^1, \varsigma^2, \dots, \varsigma^{670})$ as our experimental benchmark. For each video, corresponding captions $(s^1, s^2, \dots, s^{670})$ are generated using a video captioning model p_θ. Each caption s^i is then used as a query to calculate its conditional probabilities with all videos $(p_\theta(s^i|\varsigma^1), p_\theta(s^i|\varsigma^2), \dots, p_\theta(s^i|\varsigma^{670}))$. Based on these probabilities, a ranked list of videos is constructed for each query. The self-retrieval top-k recall, which represents the proportion of videos appearing within the top-k positions of their corresponding ranked lists, is then computed. This metric effectively measures the model's ability to capture the distinctiveness of captions [19]. As shown in Table 7, we evaluate the impact of different loss functions on the distinctive power of the model using the self-retrieval top-k recall method. The results confirm that the self-retrieval recall rates are positively correlated with the performance of video captioning models on the CIDEr metric. This positive correlation clearly highlights the importance of distinctiveness in enhancing captioning performance [12]. Experimental results demonstrate that our method design improves the model's distinctive ability, thereby enhancing the distinctiveness of generated captions.

4.5. Model efficiency

To compare the efficiency of our method with the baseline [10], an analysis was conducted on the MSVD dataset using a single RTX 3090 GPU, and three metrics were reported: the total number of model parameters (#Params(M)), training time (hours), the average time to generate a caption without mini-batching (Latency(ms)), forward FLOPs per sample without mini-batching (FLOPs (M)) and CIDEr. The results are presented in Table 8. As shown, the number of parameters increased by only 31.5M, primarily due to our weight sharing triamese decoders, with the ATF module contributing only a small number of additional parameters. The increase in training time (+1.51h) can be attributed to the requirement of hierarchal contrastive learning. When trained on larger datasets, the training time scales approximately linearly with the dataset size. However, there is no significant increase in latency, as the triamese decoders, which share weights during training, requires only sentence-level decoding during inference, thus not adding to latency. In summary, our method enhances captioning through hierarchical contrastive learning with an acceptable trade-off in terms of training cost and parameters, without introducing any additional inference overhead.

4.6. Qualitative analysis

1) Visualization: To better understand the effectiveness of our method, t-SNE [67] was used to visualize the embeddings of video captions corresponding to ten videos from the MSVD dataset for both the baseline [10] and our model, as shown in Fig. 5. The results clearly demonstrate that our approach generates embeddings with better separation, ensuring that captions from different videos are distinctively represented without coupling. Further validation was performed by visualizing the 3D t-SNE caption embeddings, as illustrated in Fig. 6. Our method results in a higher density of captions (represented in dark red) mapped within unit areas. Additionally, the scatter plots generated by our method exhibit greater dispersion, with larger coordinate values, indicating a more spread-out distribution of caption data points from different videos. Overall, our method brings the captions of the same video closer in the embedding space while creating larger separations between captions of different videos, enhancing distinctiveness. This demonstrates the feasibility and effectiveness of our approach.

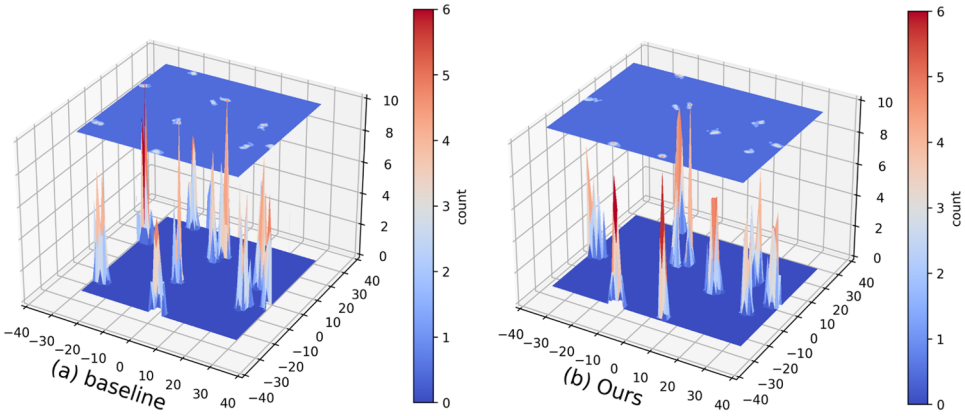

Fig. 6. 3D visualization of text embeddings produced by (a) baseline and (b) our UHCL.

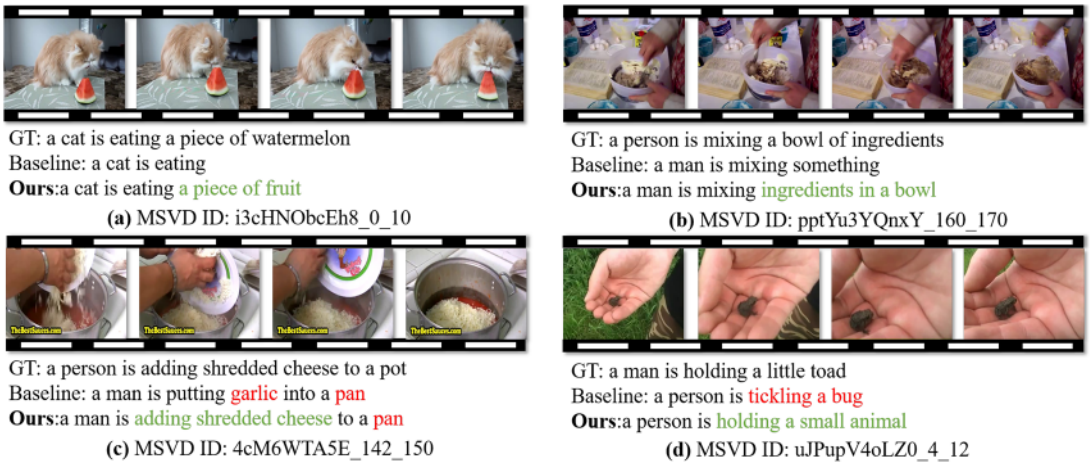

Fig. 7. Captioning examples from MSVD. Each case includes a manually annotated GT caption, captions generated by baseline and the proposed UTCL models. Accurate keywords and errors in generated captions are highlighted .

Table 7
Ablation study on self-retrieval under different loss compositions.

Exp.	Model	Self-retrieval top-k recall					Captioning
		1	5	10	20	50	CIDEr
1	w/o $\mathcal{L}_{CL}^{V}$	0.436	0.791	0.888	0.961	0.988	112.6
2	w/o $\mathcal{L}_{CL}^{N}$	0.464	0.797	0.899	0.964	0.993	113.0
3	Ours	**0.464**	**0.804**	**0.900**	**0.967**	**0.993**	**114.6**

2) Case Study: As shown in Fig. 7, four caption examples from the MSVD test set are used to compare our UHCL with the baseline [10]. Two advantages can be observed from the figure: **(1)** Our method excels at generating more accurate, fine-grained, and distinctive captions related to video scenes, possibly due to our designed hierarchical contrastive learning approach. For instance, UHCL accurately describes "fruit" in (a) and "bowl" and "ingredient" in (b). **(2)** Our UHCL performs better by producing fewer errors. The baseline misidentifies "cheese" as "garlic" in (c) and "toad" as "bug" in (d), whereas UHCL correctly identifies and generates nouns and verbs, such as "adding shredded cheese" in (c) and "holding a small animal" in (d), demonstrating the positive impact of hierarchical contrastive learning on constraining nouns and verbs.

Fig. 8 presents eight caption examples from the MSR-VTT test set, comparing our proposed method UHCL with the baseline (Fig. 8(a)-(d)) and an ablation study on hierarchical contrastive learning (Fig. 8(e)-(h)). Compared to the baseline, our method shows: **(1)** fewer errors, as seen in (a) where a man is "talking about a product" rather than "holding a box", and in (c) where our method accurately predicts "a group of people are standing." **(2)** More distinctive, as in (b) where our method correctly predicts "giving a tutorial", closely matching the ground truth "gives instructions", and in (d) where our method correctly identifies the game video, consistent with the ground truth. An ablation study on hierarchical contrastive learning reveals two key advantages of our approach: **(1)** It enriches and provides a more distinctive caption of actions in videos, facilitated by contrastive learning at the verb level. For instance, in (e)-(f), compared to single action descriptions without verb-level constraints (e.g., "talking about"), our method accurately describes actions in the video, such as "demonstrating" and "showing." **(2)** It enriches and accurately describes nouns in videos, enabled by contrastive learning at the noun-level. For example, our method accurately predicts the "interior of car" and "slam dunk" in video (g), whereas without such supervision, video captions lack distinctiveness and fail to highlight key points.

4.7. Pseudo-positives for single caption scenarios

UHCL requires positive samples. However, in some scenarios or datasets, only one caption may be available. This raises the question

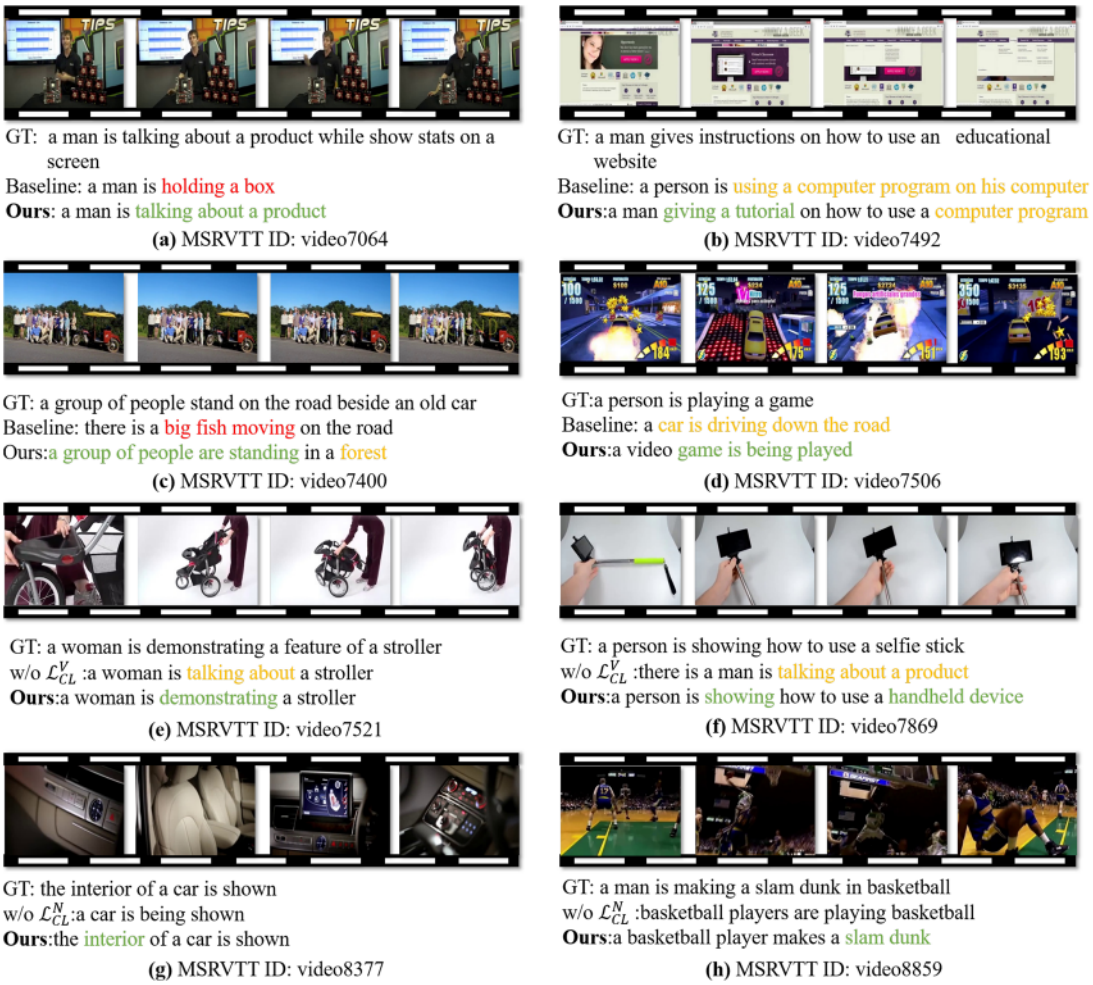

Fig. 8. Qualitative results for ablation study on MSR-VTT dataset. Accurate keywords, errors and might-be-correct in generated captions are highlighted.

Table 8
Efficiency comparison on MSVD. #Params: The toal number of model parameters. Training time: hours. Latency: The average time to generate a caption without mini-batching. FLOPs (M): Forward FLOPs per sample without mini-batching during training.

Exp.	Model	#Params(M)	Training time(h)	Latency(ms)	FLOPs(M)	CIDEr
1	Baseline	225.8	0.54	19.31	291.57	104.9
2	Ours	257.3 (+31.5)	2.05 (+1.51)	19.34 (+0.03)	583.14	114.6

Table 9
Performance comparison with the use of pseudo-positives.

Exp.	Method	MSVD			
		B@4 ↑	R ↑	M ↑	C ↑
1	Baseline	56.3	75.9	38.7	104.9
2	Baseline (Single_GT)	36.4	65.5	30.9	70.4
3	Baseline (Pseudo)	48.9	72.5	36.0	93.2
4	Ours	59.3	77.5	40.5	114.6
5	Ours (Pseudo)	53.0	74.1	36.7	101.6

of how the absence of positive samples affects the proposed method. To investigate this issue, we assume that only one ground truth is available per sample in MSVD and generate pseudo-positive samples using a large language model (GPT-5). The performance of our method based on these pseudo-positive samples is evaluated, with results presented in Table 9. Both "Baseline" and "Ours" used the original full training set for training. "Baseline (Single_GT)" refers to training each video with only one GT, while "Baseline (Pseudo)" refers to training each video with one GT along with nine pseudo-captions generated by LLM. "Ours (Pseudo)" refers to UHCL training each video with one GT along with nine pseudo-captions generated by LLM as positives.

The results indicate that performance with pseudo-positive samples is lower than with ground-truth positives for both the baseline and our method. However, our method still outperforms the baseline when no ground-truth positive samples are used. These findings show that our method still works by leveraging pseudo-positive samples, even in the absence of true positives.

4.8. Limitations

One limitation of our model is its inability to infer additional contextual information about video content, only capturing direct information from the video, as shown in Fig. 9(a)-(b). Another challenge lies in accurately predicting identities and specific scene types due to the absence of priors. This is evident in Fig. 9(c)-(d), where our model fails to pre-

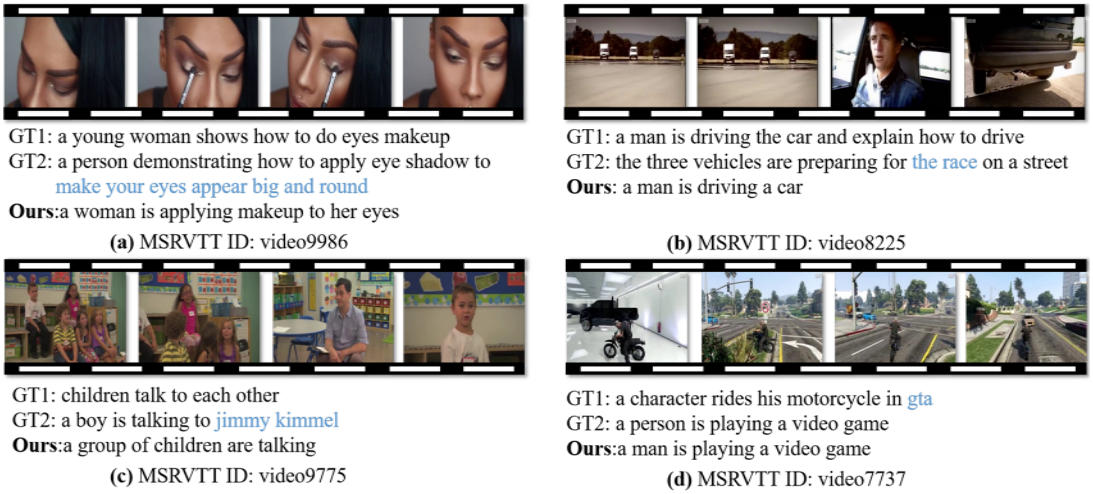

Fig. 9. Limitations of the proposed model. Each case includes two manually annotated GT captions, caption generated by the proposed UHCL models. Limitations compared to ground-truth captions are highlighted.

dict celebrity names or game titles, which impacts performance. Our method only processes visual information without leveraging external knowledge, thus, UHCL's predictions appear reasonable when considering only visual appearance.

5. Conclusion

In this paper, we propose a novel, simple yet effective unified hierarchical contrastive learning framework aimed at enhance the distinctiveness and quality of generated captions. This framework leverages triamese decoders to implement contrastive learning within a unified structure, effectively avoiding the reliance on and limitations of reference models. Furthermore, we introduce an adaptive token fusion module that enforces fine-grained, multi-level contrastive learning constraints. These advancements collectively contribute to significantly improving the quality and distinctiveness of the generated captions. Extensive experiments conducted on the widely adopted MSVD and MSR-VTT benchmarks demonstrate the effectiveness of the proposed method. In future work, we will explore the integration of multimodal knowledge graph and causal inference into multimodal contrastive learning for video captioning and other computer vision tasks.

CRediT authorship contribution statement

Haoying Sun: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Shuyi Li:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Zeyu Xi:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis, Conceptualization; **Lifang Wu:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Funding acquisition, Conceptualization.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Lifang Wu reports financial support was provided by National Natural Science Foundation of China. Shuyi Li reports financial support was provided by National Natural Science Foundation of China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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